Pervasive Computing Project Proposal

Team members

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Challenge Statement

The challenge is to design a real-world version of the “Gandalf Staff” that can guide people in their environment by lighting the way forward. In the fantasy world, the staff works through magic, but in reality we must create a framework and system that achieves similar guiding functions through technology. To keep the project feasible, the solution must balance ambition with the limits of budget and timeframe, and the problem can be viewed across three areas: hardware, software, and enclosure design, which together form the foundation for reimagining Gandalf’s staff in a modern context.

Project Idea

Building on these challenges, our project is to create a “Gandalf Staff” reimagined as a smart walking companion. It blends safety, accessibility, and a touch of magic through sensors, voice interaction, and adaptive feedback. The staff lights up automatically in the dark, with the glow symbolising Narya, the Ring of Fire entrusted to Gandalf. Inspired by its lore of hope and resilience, we reinterpret the staff as a symbol of guidance and support. It also responds to simple spoken commands to help users navigate back home or to a chosen location.

To achieve this, we will integrate:

- **Hardware**: LEDs for illumination, a microphone and speaker for interaction, haptic motors for feedback, a GPS module for navigation, and a battery-supported microcontroller.
- **Software**: Logic for data handling and interaction, supported by APIs such as Google Maps for navigation and Gemini for voice control.
- **Enclosure**: A custom design resembling the original staff, built with 3D printing and laser cutting to balance aesthetics, durability, portability, and safety.

This reimagining captures the “magic” of Gandalf’s staff through technology, making it both a practical mobility aid and an inspiring companion.

User Journey

- **Context**: The user carries the staff as both a mobility aid and smart assistant.
- **Trigger**: The staff lights up automatically in the dark or responds to a spoken command such as “Go back home.”
- **Steps**:
    - The staff glows to provide illumination.
    - The microphone captures the user’s voice command.

- The system processes the speech, connects to a navigation service, and interprets the request.
    - The speaker gives clear step-by-step directions.
    - The haptic system provides gentle vibrations when the user heads in the wrong direction.
- **After interaction**: Once the destination is reached, the staff returns to standby.

Parts Required

Hardware
- **Core system**: Microcontroller, battery, connection cables, and power module.
- **Sensing and interaction**: Light sensor, microphone, speaker, haptic motors, and GPS module (e.g., NEO-6M).
- **Lighting**: LED or bulb for illumination.
- **Enclosure**: Wooden or metal staff body with a 3D-printed case to house electronics.

Software
- **APIs and services**: Google Maps (for navigation), Gemini (for AI/voice interaction).
- **Design tools**: Fusion 360 for 3D modelling and enclosure design.

Timeline

Task	**Who?**	**Week**									
		4	5	6	7	8	9	10	11	12	13
Research	All										
Finalise the idea concept	All										
Hardware Development	Sydney, Ford										
Software Development	Kiba, Yu										
Documenting	All										
Enclosure Printing	All										
Component Integrations	All										
Prototype Testing	All										

Potential Risks

Our team has more experience in software than in hardware or 3D printing, so we may need extra time to learn how to design, print, and assemble the enclosure. To manage this, we plan to start with simple prototypes and use online resources to build our skills early in the project.

Another risk is the availability of 3D printing and laser cutting facilities. If machines are busy or scheduling conflicts arise, the timeline could be affected. We will reduce this by booking equipment in advance, allowing buffer time, and preparing smaller test prints before the final design.

Finally, some hardware parts may not be available in labs or local stores, which could add cost and delay if we need to order them. Compatibility issues may also arise when combining sensors, microcontrollers, power supply, and the enclosure. To prepare, we will identify alternative suppliers, keep backup components, and test modules individually before full integration. These risks are manageable and will help us grow as a team.

Appendix

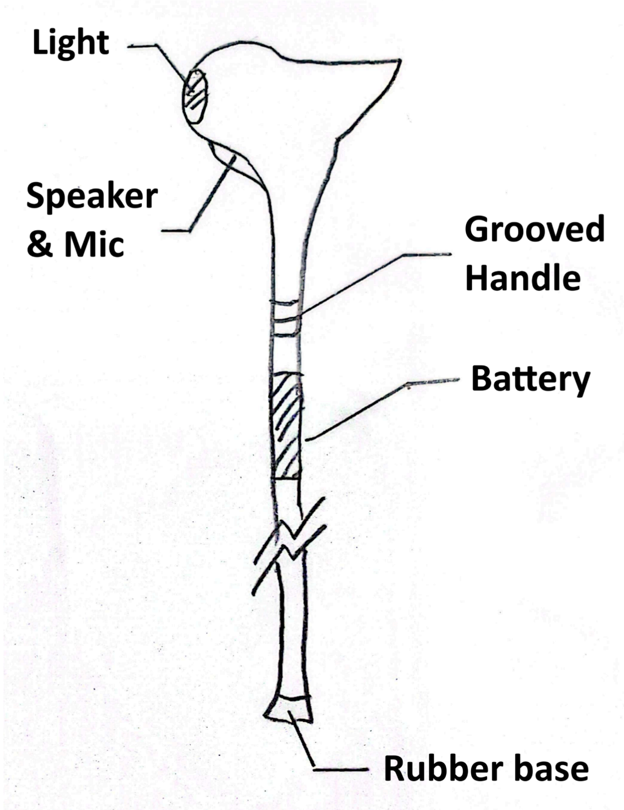

Fig 1: The Initial Sketch of the Proposal

Fig 2: The Commercial Hiking Stick

Fig 3: The Gandalf Staff from the Lord of the Rings